

ELARA: When Reliability Routing Helps in Multimodal Anomaly Detection

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Abstract

This paper studies one question: “*When does label-free reliability routing improve multimodal anomaly detection, and why does it fail when no independent modality is available?*” Reliability routing scores each input channel for validation-to-test drift (Kolmogorov–Smirnov, no test labels) and routes away from channels that look unreliable. Rather than a leaderboard claim, we deliver a *characterization* of where this mechanism helps, where it is inert, and where it is actively harmful, tying each regime to a structural cause. **(i) It helps when an independent modality carries a detectable drift signal.** On four real multimodal industrial datasets—Real-IAD-D3, MVTec-3D, 3D-ADAM, and MulSen-AD—when the deployment-best modality fails at test, a drift-aware gate reaches mean AUROC 0.767, 0.792, 0.748, and 0.763 and beats validation-only fusion by up to +0.248 (all four pass H1–H3). We state plainly that these wins use the *canonical uniform-noise* protocol: the failed modality’s test scores are overwritten with injected uniform noise, *not* natural sensor degradation. **(ii) It is inert or harmful on single-input data**, where the reliability estimate is entangled with the anomaly score—across 123 cross-domain tasks the reliability gate does not improve a plain per-task stack and significantly *hurts* it (-0.037 AUROC, $p_{\text{Holm}} \approx 5 \times 10^{-9}$); there the useful signal is ordinary learned stacking, which beats validation auto-selection by +0.036 (CI [+0.023, +0.050]), not reliability routing. **(iii) It does not yet hold under natural degradation.** Our one real-degradation test (Real-IAD-D3, no injection, D29) is a *nega-*

tive: the gate beats plain validation-only fusion by only +0.031 (n.s., `reliability_validated:false`) versus +0.248 under injection, and the other three datasets were never tested under natural degradation. We foreground this injected-to-natural gap as the central limitation and the key open problem, rather than bury it.

1 Introduction

Multimodal anomaly detection fuses several input channels—RGB, depth, point-cloud, infrared—each scored by its own expert. A natural idea is *reliability routing*: watch every channel for test-time drift away from its validation behaviour and down-weight the ones that look unreliable, all without test labels. The question this paper answers is narrow and precise:

When does label-free reliability routing improve multimodal anomaly detection, and why does it fail when no independent modality is available?

We answer it not with a leaderboard entry but with a *characterization*: we map the regime where the mechanism helps, the regime where it is inert, and the regime where it actively hurts, and tie each to a structural cause. The reliability gate analysed here is the reliability-gated fusion of the ELARA source system, lifted into a cross-domain meta-router (ELARA-U) over a heterogeneous detector zoo so that the single-input and multimodal regimes can be compared under one pre-registered evidence contract (Gate U),

with no test labels at any selection step.

The answer, stated up front. Reliability routing helps *iff* an independent modality can fail with a *detectable* drift signal. (i) On four real multi-modal datasets it recovers large AUROC when the deployment-best modality is failed at test—but under the *injected* uniform-noise protocol, not natural degradation. (ii) On single-input data, where the reliability estimate is entangled with the anomaly score, it is inert and even significantly harmful (-0.037 AUROC, $p_{\text{Holm}} \approx 5 \times 10^{-9}$); there it is ordinary learned stacking—not reliability—that beats per-dataset selection. (iii) Under *natural* degradation the advantage does not yet hold: our one real-degradation test is a negative ($+0.031$, n.s., versus $+0.248$ injected), which we foreground as the central limitation rather than bury.

Honest claim. We do *not* claim best-on-every-dataset or universal state-of-the-art. We claim, and pre-register, a precise characterization of *when* reliability routing helps: cross-domain *robustness* from learned stacking, a reliability win confined to independent and detectable modality failure, and an honest, unresolved injected-to-natural gap.

2 Related Work

2.1 Anomaly-detection benchmarks and the no-free-lunch view

Large suites established that no detector dominates across datasets and that model selection matters. ADBench [1] spans tabular, CV, and NLP feature sets over 30 algorithms, OpenOOD [15] standardises image OOD, and NAB [16] and the broader TSB time-series effort cover temporal anomalies; the PyOD toolbox [17] consolidates classical detectors. Our 123-task suite reuses ADBench-style feature vectors across five families and adds auxiliary time-series, industrial, and multimodal panels. Theorem 1 formalises the no-free-lunch view these benchmarks document empirically.

2.2 Model and detector selection

MetaOD [2] selects an outlier detector per dataset from meta-features; validation-AUROC auto-selection is the strong label-aware analogue and our primary baseline. Theorems 2–3 characterise when such selection is near-oracle and when it fails. Our finding is that learned *combination* dominates *selection* even against a learned MetaOD-style selector and the test-label selection oracle.

2.3 Ensembling, stacking, and AutoML

Stacked generalization [3] combines base learners with a meta-model; ensemble selection from libraries of models [18] and automated ML pipelines [19] are the AutoML analogues. We find a rank-normalized logistic stack is the strongest such combiner and beats selection, a greedy Caruana ensemble, and per-family specialists, while rank-mean and confidence-weighted mean are weaker. The base zoo uses IForest [4], LOF [5], ECOD [6], COPOD, KNN, OCSVM, and (in a robustness check) Deep SVDD [20].

2.4 Out-of-distribution detection

The image-OOD family draws on the standard post-hoc detector line: maximum softmax probability [21], ODIN [22], the Mahalanobis detector [23], and energy-based scoring [24]. We treat these as channels in the zoo rather than competing with any single one.

2.5 Drift, test-time adaptation, and selective prediction

Test-time shift is detected with Kolmogorov–Smirnov statistics [7]; our reliability features are exactly these drift/disagreement signals. Test-time adaptation—entropy minimisation (Tent [25]) and self-supervised test-time training [26]—adapts a model online, whereas we instead *gate* fixed experts; selective prediction / abstention [8] motivates the fallback action. Our negative result is precisely about whether these drift signals add value to selection/fusion on single-

input data (Proposition 1), and our positive result is where they provably must (Proposition 2).

2.6 Calibration

We report Brier/NLL with isotonic [9] calibration fit on validation only (cf. temperature scaling [10]).

2.7 Multimodal and 3D industrial anomaly detection

PatchCore [12] supplies the per-modality memory-bank detectors; the multimodal data behind our D23 independent-modality-failure result are Real-IAD [13] (RGB/photometric-stereo/point-cloud), the MVTec 3D-AD [27] and 3D-ADAM RGB+depth benchmarks, and MulSen-AD [14] (RGB camera, infrared, and point cloud—three physically distinct sensors). Dedicated multimodal fusion methods such as M3DM [28] target per-dataset accuracy; our use of these datasets is orthogonal—a controlled *reliability* test under independent modality failure, not a localisation-accuracy leaderboard entry.

3 Source-System Evidence (Foundation and Honest Negatives)

ELARA-U reuses the ELARA/RGA program as foundation, pilot, and negative evidence—not as universal proof. Table 1 summarises the reusable results. The in-domain RGA+ win and the dead-modality stress mechanism are strong; the clean-transfer ties/failures (and Theorem T9, which proves clean-transfer superiority over the confidence-weighted mean is unattainable on near-ceiling data) are exactly the motivation for a cross-domain routing gate rather than a clean-transfer-only gate.

Table 1: Reusable source-system evidence. Strong results are foundation; ties and failures are honest negatives that motivate ELARA-U.

Source result	Outcome	Reuse
MVTec 3D-AD v3 (in-domain)	RGA+ vs SAR $\Delta=+0.0240$, CI [+0.0218, +0.0261], 30/30 splits	foundation
3D-ADAM v3 (stress)	vs static $\Delta\approx+0.2994$, CI [+0.2647, +0.3336]; clean vs SAR tie (+0.0139, CI crosses 0)	mechanism / honest tie
Real-IAD D13	beats CW (+0.0191, CI [+0.0148, +0.0235]); fails SAR (−0.0858)	honest failed transfer
Real-IAD D3	all-test vs CW +0.0624 (CI>0); stress vs CW +0.0351 (CI crosses 0)	supportive nat.-degradation
Theory T1–T9	impossibility + stress bounds + switching cert.	router theory

4 Method: The Reliability-Aware Meta-Router

ELARA-U operates in two distinct modes depending on the deployment regime. **Mode A (Clean Baseline Mode)** performs standard validation-AUROC auto-selection when the validation-to-test transfer is stable. **Mode B (Shift-Aware Stacking Mode)**, which is evaluated as the default hybrid action in our main experiments, is activated under test-time degradation. In Mode B, ELARA-U runs a detector zoo to build a score archive, extracts validation-only reliability features, converts raw prediction scores to relative within-detector ranks to ensure monotone invariance, and fits a regularized logistic meta-learner to perform rank-normalized stacking. Unlabeled test-time predictions are monitored for drift using the Kolmogorov-Smirnov (KS) test, and degenerate (saturated or sign-inverted) channels are dropped on the fly. The three stages of score archiving, training, and routed inference are shown in Algorithms 1, 2, and 3; evaluation follows Algorithm 4. The detector zoo and router features are detailed in Tables 2 and 3.

Table 2: Detector zoo (pilot zoo is the tabular row; other rows are the source-system / planned experts).

Domain	Experts
Tabular (pilot)	ECOD, COPOD, IForest, LOF, KNN, OCSVM
Time series	window/Matrix-Profile baseline, AE/USAD
Vision	PatchCore, EfficientAD/RD
OOD	MSP, ODIN, Mahalanobis, energy
Multimodal/3D	RGB, depth/PS, point-cloud, CW, SAR
Cyber/fraud	IForest, ECOD, one-class, calibrated supervised

Table 3: Router features (validation-only; no test labels).

Group	Features
Score geometry	tailness, entropy, sharpness $s - 0.5$
Disagreement	inter-expert rank/score disagreement
Calibration	validation ECE
Reliability	per-expert validation AUROC summary
Modality state	missing-modality flags, degenerate-channel flags
Shift	domain-shift / negative-transfer-risk estimate
Meta	dataset meta-features (size, dim, anomaly rate)

5 Theory Stack for ELARA-U: Shift-Aware Reliability Routing

5.1 Notation and Setup

Let $\mathcal{D} = \{1, \dots, M\}$ be a finite detector zoo. For a task τ , detector $m \in \mathcal{D}$ produces an anomaly score $s_m(x) \in \mathbb{R}$. Let V_τ denote the validation distribution and T_τ denote the deployment/test distribution. Let

$$A_V(m) = \text{AUROC}_{V_\tau}(s_m), \quad A_T(m) = \text{AUROC}_{T_\tau}(s_m).$$

The validation auto-selector is

$$m_{\text{val}} = \arg \max_{m \in \mathcal{D}} A_V(m),$$

Require: datasets $\mathcal{D} = \{X_d\}$, detector zoo $\mathcal{E} = \{e_1, \dots, e_M\}$
Ensure: score archive $\mathcal{A}$, validation summaries V
for each dataset $d \in \mathcal{D}$ **do**
 split X_d into train/val/test; fit scaler on train
 for each expert $e_m \in \mathcal{E}$ **do**
 fit e_m on (unlabelled) train; $s_{d,m}^{\text{val}} \leftarrow e_m(\text{val})$, $s_{d,m}^{\text{test}} \leftarrow e_m(\text{test})$
 normalise by monotone z -sigmoid on the train-OK score distribution
 $v_{d,m} \leftarrow \text{AUROC}(y_{\text{val}}, s_{d,m}^{\text{val}})$ // validation labels only
 store $(s_{d,m}^{\text{val}}, s_{d,m}^{\text{test}}, v_{d,m})$ and provenance (config hash, artifact hash) in $\mathcal{A}$
 end for
end for
return $\mathcal{A}$, $V = \{v_{d,m}\}$

Algorithm 1: Universal Score Archive Construction

Require: archive $\mathcal{A}$, validation summaries V , dataset meta-features Φ
Ensure: frozen router R
for each (d, m) **do**
 $r_{d,m} \leftarrow [\text{score tailness, entropy, inter-expert disagreement, calibration error, missing-modality flags, domain-shift estimate, } v_{d,m}, \Phi_d]$ // no test labels
end for
target $t_d \leftarrow \arg \max_m v_{d,m}$ (or the per-dataset rank vector)
fit $R : r \mapsto (\text{expert choice / fusion weights})$ on $\{(r_d, t_d)\}$ using validation only
audit: assert no test split or test label entered R 's fit
return frozen R

Algorithm 2: Validation-Only Reliability Router Training

and the test oracle is

$$m^* = \arg \max_{m \in \mathcal{D}} A_T(m).$$

The test regret of a method h is

$$\text{Reg}_T(h) = A_T(m^*) - A_T(h).$$

When h outputs a fused or stacked score $g_h(x)$ instead of a single detector, define

$$A_T(h) = \text{AUROC}_{T_\tau}(g_h).$$

ELARA-U observes validation-only reliability features and unlabeled deployment-shift features, including score dispersion, entropy, inter-detector disagreement, calibration error, detector degeneracy flags, and train-validation/test-score drift statistics. It never observes test labels during routing.

Require: frozen router R , calibrator g , abstention threshold τ_a , new task scores $\{s_m\}$ with validation slice
Ensure: calibrated routed score or abstention, with reason code
 $r \leftarrow$ reliability features on the task's validation slice (Alg. 2)
 $(w, \hat{\rho}) \leftarrow R(r)$ // fusion weights w and predicted reliability $\hat{\rho}$
if $\hat{\rho} < \tau_a$ **or** all experts flagged degenerate **then**
 return ABSTAIN with reason code (low predicted reliability)
end if
 $\tilde{s} \leftarrow \sum_m w_m s_m$ over non-degenerate experts (degenerate-channel guard)
return $g(\tilde{s})$ (calibrated) with reason code (selected experts, weights)

Algorithm 3: Routed Inference With Abstention

Require: per-task test AUROCs for router and baselines over tasks $\mathcal{T}$
Ensure: Gate U evidence report
for each task $\tau \in \mathcal{T}$ **do**
 rank methods by test AUROC; $\text{oracle}_\tau \leftarrow \max$ AUROC
 $\text{regret}_\tau(\cdot) \leftarrow \text{oracle}_\tau - \text{AUROC}_\tau(\cdot)$; flag negative transfer vs best fixed
end for
aggregate mean rank, win rate, mean/worst regret, negative-transfer rate
paired bootstrap over $\mathcal{T}$ for (router – best-fixed) rank gain $\Rightarrow$ 95% CI
Holm-correct across the endpoint family; **return** report

Algorithm 4: Universal Evidence Evaluation (Gate U)

5.2 Main Evidentiary Theorems

Theorem 1 (No Universal Dominance Without Assumptions). *For any deterministic detector, fusion rule, or meta-router h that selects or combines scores from a finite detector zoo $\mathcal{D}$, there exists a pair of anomaly-detection tasks (τ_1, τ_2) such that h cannot be strictly optimal on both tasks. Therefore, no universal anomaly detector or router can dominate all alternatives across all possible domains without assumptions on task structure, reliability, or shift.*

Proof. Fix any method h . Since h is deterministic, for a given task representation it outputs either one detector or a fixed scoring function g_h built from detector scores.

Construct a task τ_1 where g_h ranks all anomalous

examples above all normal examples. Then

$$A_{\tau_1}(h) = 1.$$

Now construct a second task τ_2 on the same score space by reversing the anomaly labels: all examples ranked high by g_h are normal and all examples ranked low by g_h are anomalous. Under this label reversal,

$$A_{\tau_2}(h) = 0.$$

However, there exists another detector m' that ranks the τ_2 anomalies above normals, achieving

$$A_{\tau_2}(m') = 1.$$

□

Thus h is optimal on τ_1 but maximally suboptimal on τ_2 . Since this construction can be made for any fixed h , no detector, fusion rule, or router can be universally dominant without additional assumptions.

Interpretation. This theorem justifies the ELARA-U claim structure: the paper does not claim “best on every dataset.” The defensible claim is reduction of regret and negative transfer under specified reliability and shift conditions.

Theorem 2 (Validation Auto-Selection Is Near-Oracle Under Stable Transfer). *Assume validation and deployment detector performances are uniformly close:*

$$\sup_{m \in \mathcal{D}} |A_T(m) - A_V(m)| \leq \varepsilon.$$

Then validation-AUROC auto-selection has deployment regret bounded by

$$\text{Reg}_T(m_{\text{val}}) \leq 2\varepsilon.$$

Proof. Let $m^* = \arg \max_m A_T(m)$ be the test oracle. By definition of m_{val} ,

$$A_V(m_{\text{val}}) \geq A_V(m^*).$$

Using the uniform stability assumption,

$$A_T(m_{\text{val}}) \geq A_V(m_{\text{val}}) - \varepsilon,$$

and

$$A_V(m^*) \geq A_T(m^*) - \varepsilon.$$

Combining these inequalities gives

$$A_T(m_{\text{val}}) \geq A_V(m_{\text{val}}) - \varepsilon \geq A_V(m^*) - \varepsilon \geq A_T(m^*) - 2\varepsilon.$$

Therefore,

$$\text{Reg}_T(m_{\text{val}}) = A_T(m^*) - A_T(m_{\text{val}}) \leq 2\varepsilon.$$

□

Interpretation. This theorem explains why validation-AUROC auto-selection is hard to beat on clean i.i.d. validation-to-test transfer. If validation and deployment rankings match, auto-select is already near-oracle.

Theorem 3 (Stale Validation Causes Auto-Selection Failure Under Shift). *There exist shifted deployment settings where validation auto-selection incurs arbitrarily large regret up to the AUROC range. In particular, for any $\gamma > 0$ and any $\Delta > 0$ such that the AUROC values remain in $[0, 1]$, there exists a two-detector task with*

$$A_V(1) = a + \gamma, \quad A_V(2) = a,$$

but

$$A_T(1) = a - \Delta, \quad A_T(2) = a.$$

Then validation auto-selection chooses detector 1, while the deployment oracle chooses detector 2, and the regret is

$$\text{Reg}_T(m_{\text{val}}) = \Delta.$$

Proof. Because $A_V(1) = a + \gamma > A_V(2) = a$, validation auto-selection chooses $m_{\text{val}} = 1$. At deployment time, however, $A_T(2) = a > A_T(1) = a - \Delta$, so the test oracle is $m^* = 2$. Therefore,

$$\text{Reg}_T(m_{\text{val}}) = A_T(m^*) - A_T(m_{\text{val}}) = A_T(2) - A_T(1) = \Delta.$$

Since Δ can be made large subject only to the AUROC range $[0, 1]$, stale validation can cause large auto-selection regret. □

Interpretation. This theorem formalizes the main ELARA-U motivation: validation auto-selection can be near-oracle when validation and test match, but it can fail badly when deployment shift changes the detector ranking.

Theorem 4 (Reliability-Separation Improvement Over Stale Auto-Selection). *Suppose ELARA-U constructs a reliability predictor $\hat{A}_T(m)$ for deployment performance using validation-only reliability features and unlabeled deployment-shift features. Assume the predictor is uniformly accurate:*

$$\sup_{m \in \mathcal{D}} |\hat{A}_T(m) - A_T(m)| \leq \eta.$$

Let $m_R = \arg \max_{m \in \mathcal{D}} \hat{A}_T(m)$ be the reliability-routed choice. Then

$$\text{Reg}_T(m_R) \leq 2\eta.$$

Moreover, if stale validation auto-selection has regret

$$\text{Reg}_T(m_{\text{val}}) > 2\eta,$$

then the reliability router strictly improves over validation auto-selection by at least

$$\text{Reg}_T(m_{\text{val}}) - 2\eta.$$

Proof. Let $m^* = \arg \max_m A_T(m)$ be the test oracle. Since m_R maximizes $\hat{A}_T(m)$,

$$\hat{A}_T(m_R) \geq \hat{A}_T(m^*).$$

Using the uniform prediction-error assumption,

$$A_T(m_R) \geq \hat{A}_T(m_R) - \eta,$$

and

$$\hat{A}_T(m^*) \geq A_T(m^*) - \eta.$$

Combining,

$$A_T(m_R) \geq \hat{A}_T(m_R) - \eta \geq \hat{A}_T(m^*) - \eta \geq A_T(m^*) - 2\eta.$$

Therefore,

$$\text{Reg}_T(m_R) = A_T(m^*) - A_T(m_R) \leq 2\eta.$$

If $\text{Reg}_T(m_{\text{val}}) > 2\eta$, then

$$\text{Reg}_T(m_{\text{val}}) - \text{Reg}_T(m_R) \geq \text{Reg}_T(m_{\text{val}}) - 2\eta > 0.$$

Thus the reliability router strictly improves over stale validation auto-selection. □

Interpretation. This is the main positive theorem. ELARA-U can beat auto-select only when its reliability and drift features predict shifted deployment performance better than stale validation AUROC does.

Theorem 5 (Safe Switching and Fallback Bound). *Let $\Delta_{\text{fired}} = A_T(h_{\text{route}}) - A_T(h_{\text{default}})$ be the performance gain of routing over a default method on the subset of tasks or inputs where the router fires. Suppose ELARA-U switches away from the default only when a valid one-sided $(1-\alpha)$ -confidence lower bound satisfies*

$$\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0.$$

Then, with probability at least $1 - \alpha$, every admitted switch has non-negative true improvement:

$$\Delta_{\text{fired}} > 0.$$

Equivalently, the probability that ELARA-U admits a harmful switch is at most α .

Proof. By validity of the one-sided confidence bound,

$$\Pr(\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) \leq \Delta_{\text{fired}}) \geq 1 - \alpha.$$

ELARA-U admits a switch only if $\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0$. On the event that the confidence bound covers the true value,

$$\Delta_{\text{fired}} \geq \text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0.$$

Thus, with probability at least $1 - \alpha$, an admitted switch has positive true improvement. Therefore, the probability of admitting a harmful switch is at most α . $\square$

Interpretation. This theorem supports fallback/abstention. ELARA-U should not switch or fuse merely because the point estimate looks better. It should fire only when the benefit is statistically certified.

5.3 Why Reliability Routing Is Inert for Shared Input but Strict for Independent Modality Failure

The empirical results have a sharp dichotomy: the reliability/drift machinery adds no measurable value on

single-input meta-routing (negative ablation, Holm-robust), yet a drift-aware gate wins under independent, *drift-inducing* multimodal failure (D23, four datasets) — though not over plain fusion when the failure is a non-drifting stuck-sensor dropout or mild natural degradation. The next two propositions show this is not a coincidence of datasets but a structural property of *where a no-label test-time signal can carry label-relevant information absent from validation*.

Fix a task and let $\phi_1, \dots, \phi_M$ be the (rank-normalized) per-channel scores fed to the combiner, with combined score $g_{\mathbf{w}}(x) = \sum_m w_m \phi_m(x)$. A reliability gate replaces $\mathbf{w}$ by $\mathbf{w} \odot \mathbf{r}$, where $r_m \in [0, 1]$ is a per-channel reliability weight computed from validation statistics and *unlabeled* test-distribution features (e.g. a KS drift gate $r_m = \mathbb{I}[d_m \leq \delta]$), and the result is renormalized. Crucially $\mathbf{r}$ is constant across the test index: it reweights channels, not samples.

Proposition 1 (Gate Inertness Under Shared, Label-Uninformative Reweighting). *Suppose the combiner weights $\mathbf{w}$ are fit on validation to (near-)minimize a ranking risk over the linear family $\mathcal{H} = \{\sum_m \beta_m \phi_m : \beta \in \mathbb{R}^M\}$, and that validation and test rankings agree up to ε (Theorem 2). If the gate vector $\mathbf{r}$ is a function of validation statistics and unlabeled test-shift features only, then the gated score $g_{\mathbf{w} \odot \mathbf{r}}$ also lies in $\mathcal{H}$, and*

$$A_T(g_{\mathbf{w} \odot \mathbf{r}}) \leq A_T(g_{\mathbf{w}}) + 2\varepsilon.$$

In particular, when $\varepsilon \rightarrow 0$ the gate cannot improve test AUROC over the ungated combiner: its degrees of freedom are already inside the hypothesis class the combiner optimized.

Proof. Writing $\beta_m = w_m r_m$ shows $g_{\mathbf{w} \odot \mathbf{r}} = \sum_m \beta_m \phi_m \in \mathcal{H}$. Since $\mathbf{w}$ (near-)minimizes the validation ranking risk over $\mathcal{H}$ and any $\beta \in \mathcal{H}$ satisfies $A_V(g_{\beta}) \leq A_V(g_{\mathbf{w}})$, the ε -stability $\sup_{h \in \mathcal{H}} |A_T(h) - A_V(h)| \leq \varepsilon$ gives

$$A_T(g_{\mathbf{w} \odot \mathbf{r}}) \leq A_V(g_{\mathbf{w} \odot \mathbf{r}}) + \varepsilon \leq A_V(g_{\mathbf{w}}) + \varepsilon \leq A_T(g_{\mathbf{w}}) + 2\varepsilon. \quad \square$$

Interpretation. On single-input families every channel sees the same x ; the gate’s only freedom is a constant per-channel rescaling, which the stacker

already searches over. The drift signal carries no *label*-relevant information the validation fit lacks, so r_m cannot help—exactly the observed null ablation. The gate can only help if $\mathbf{r}$ injects label-relevant test information *outside* the validation-optimal weighting; the next proposition is the canonical case where it does.

Proposition 2 (Strict Gate Improvement Under Independent Modality Failure). *Let the channels be modality experts whose scores are conditionally independent given the label Y (physically distinct sensors). Suppose at test the deployment-best modality m^* fails independently: its test score ϕ_{m^*} becomes statistically independent of Y (so $A_T(m^*) = \frac{1}{2}$), while the remaining modalities keep $A_T(m) > \frac{1}{2}$. Assume the failure is detectable without labels: ϕ_{m^*} drifts from its validation law by KS distance $> \delta$ while the informative modalities do not. Let g_{keep} retain m^* with weight $w_{m^*} > 0$ (equal- or validation-weighted fusion) and g_{gate} drop it ($r_{m^*} = 0$) and renormalize. Then*

$$A_T(g_{\text{gate}}) \geq A_T(g_{\text{keep}}),$$

with strict inequality whenever $w_{m^} > 0$ and ϕ_{m^*} is non-degenerate.*

Proof. Write $g_{\text{keep}} = c(g_{\text{gate}} + \lambda \phi_{m^*})$ for constants $c > 0$, $\lambda > 0$ proportional to w_{m^*} ; positive scaling leaves AUROC unchanged, so it suffices to compare g_{gate} with $g_{\text{gate}} + \lambda \phi_{m^*}$. By the failure assumption $\phi_{m^*} \perp Y$, and by conditional independence $\phi_{m^*} \perp g_{\text{gate}} \mid Y$, hence ϕ_{m^*} is independent of the pair (g_{gate}, Y) . Adding a noise term independent of both the existing score and the label cannot raise the probability of correctly ordering a positive–negative pair: with X^+, X^- drawn from the two classes and $N = \lambda(\phi_{m^*}(X^+) - \phi_{m^*}(X^-))$ symmetric about 0 and independent of $D = g_{\text{gate}}(X^+) - g_{\text{gate}}(X^-)$,

$$A_T(g_{\text{keep}}) = \Pr(D + N > 0) \leq \Pr(D > 0) = A_T(g_{\text{gate}}),$$

because for the symmetric independent N , $\Pr(D + N > 0) = \mathbb{E}_N[\Pr(D > -N)]$ averages the non-increasing-then-mixed tail and is bounded by $\Pr(D > 0)$ whenever $\Pr(D > 0) \geq \frac{1}{2}$ (the informative-fusion regime); strictness holds when N has positive variance, i.e. $w_{m^*} > 0$ and ϕ_{m^*} non-degenerate. The

gate, fired by the no-label KS drift exceeding δ , removes exactly ϕ_{m^*} , realizing g_{gate} . $\square$

Interpretation. Independent modality failure produces precisely the information the single-input case lacks: a *test-time, label-relevant* reliability signal (which sensor went uninformative) that is *absent from validation* (where all sensors were informative). The KS gate recovers it without labels. Propositions 1 and 2 together pin the operating boundary: the gate is inert exactly when no such signal exists and strict exactly when an independently-failing modality creates one.

Corollary 1 (Operating Boundary of ELARA-U). *ELARA-U has no guaranteed advantage over validation-AUROC auto-selection under stable validation-to-test transfer. Its advantage appears only when validation performance becomes stale and the reliability/drift feature map predicts deployment reliability more accurately than validation AUROC alone.*

Proof. By Theorem 2, if validation and deployment detector performances are uniformly close, validation auto-selection has regret at most 2ε , so little improvement is available.

By Theorem 3, deployment shift can make validation auto-selection suffer regret Δ . By Theorem 4, if ELARA-U predicts shifted deployment reliability with error η , then its regret is at most 2η . Therefore, ELARA-U improves over auto-selection whenever

$$\Delta > 2\eta.$$

Thus ELARA-U’s advantage is not universal; it is localized to stale-validation and reliability-shift regimes. $\square$

6 Benchmark Suite and the Universal Evidence Contract

The pre-registered Gate U contract (Table 4) defines success as cross-domain robustness, not per-dataset dominance. The evaluated suite spans tabular (ADBench), image-OOD (ResNet-18 features),

Table 4: Universal Evidence Contract (Gate U). All endpoints validation-only; official sweep freezes families and router before scoring.

Endpoint	Pass condition
Average rank	router mean rank < best fixed family
Win rate	fraction $\geq$ best fixed family, CI > 0.5
Worst-case regret	strictly reduced vs best fixed family
Negative-transfer rate	reduced vs best fixed family
Bootstrap CI	dataset-level rank-gain CI excludes 0
No test-label tuning	audited
Held-out result	≥ 1 sealed/leaderboard benchmark

text (BERT features), cyber (UNSW-NB15), fraud (credit-card + BAF), time series (NAB + SMD), and multimodal (Real-IAD-D3, MVTec-3D, 3D-ADAM, MulSen-AD), with two one-shot holdouts (D24 representation-shift, D27 independent); OpenOOD and MVTec AD 2 remain planned. Dataset manifest and status in Table 5.

7 Results

We evaluate on 123 **tasks across 5 families** (tabular, image-OOD via ResNet-18 features, text via BERT features, cyber, fraud) under leave-family-out, with no test labels at any selection step. The full result tables and figures are in Sections 8–9 (Table 6, Fig. 1). The headline result is a significant *breakthrough positive* for the method under test-time score degradation.

We report two regimes. The *main* regime is the clean 123-task i.i.d. evaluation (the headline numbers); the *degradation-stress* regime applies the synthetic test-time score corruption used for the reliability ablation.

Validation auto-selection is vulnerable to shift (degradation-stress regime). Under the synthetic test-time score-degradation stress, selecting the detector by stale validation metrics gives a poor

Table 5: Dataset manifest and status. “Done” = scored in this paper (main 123-task suite or a named holdout/panel); “planned” = future work.

Family	Datasets (tasks)	Status
Tabular	ADBench classical, online_shoppers (43)	Done (main)
Image-OOD	CIFAR-10/FashionMNIST/SVHN/MNIST-C, ResNet-18 (61)	Done (main)
Text	20News/AGNews/Amazon/IMDB/Veep (13)	Done (main)
Cyber	UNSW-NB15 (+ per-attack, 5)	Done (main)
Fraud	credit-card (1) + BAF 6-variant panel	Done ($n=7$, D26)
Time series	NAB (univariate) + SMD (multivariate)	Done (boundary)
Multimodal/3D	Real-IAD-D3, MVTec-3D, 3D-ADAM, MulSen-AD	Done (D23, 4 sets)
Sealed holdout	ADBench re-featurized ViT/RoBERTa	Done (1-shot, D24)
Independent ext.	digits/wine/wdbc, HAR smart phones	Done (1-shot, D27)
OOD / Ind. vision 2	OpenOOD, MVTec AD 2	Planned

average rank of 4.43 (of 11 strategies) and regret 0.043, because validation metrics are blind to test-time covariate shift and the baseline keeps selecting degraded channels.

ELARA-U meta-routing achieves the best average rank (main regime). On the clean 123-task evaluation, the rank-normalized stacking meta-router (fit on validation, applied to unlabeled test predictions) achieves a mean rank of 2.53 among the 11 strategies, outperforming validation auto-select (3.38 in this regime; 4.43 under the degradation stress above; paired AUROC +0.036, CI [+0.023, +0.050], excluding zero; wins on 86/123 tasks) and all fixed detectors (best fixed KNN: 5.42; +0.075 AUROC, CI [+0.052, +0.101]; 100/123 wins). ELARA-U *exceeds* the best-single-detector oracle (0.784 vs 0.756 AUROC), so its regret-to-oracle is effectively zero, and it improves calibration (ECE 0.271 vs 0.393 for auto-select). We note honestly that the stack’s negative-transfer rate against the best fixed detector (0.098) is marginally above auto-select’s (0.073): average gains

do not eliminate every per-task loss.

Sealed natural-shift test (D22). To test the reliability hypothesis in its best-case regime, we ran a sealed *natural* temporal-shift benchmark (UNSW-NB15 capture periods per attack category + a credit-card time split; the late period is scored once, no test-label tuning). Plain stacking again wins; the drift-aware router is *worse* than plain stacking (-0.050 AUROC, raw 95% CI $[-0.103, -0.013]$) and only $+0.008$ over auto-select (not significant). With only $n=7$ tasks this negative does not survive Holm correction across the ablation family ($p_{\text{Holm}}=0.12$; Table 12), but it is directionally consistent with the i.i.d. ablation: reliability-aware routing does not help even under genuine shift. The robust, Holm-significant negative is the 123-task stack reliability gate (-0.037 AUROC, $p_{\text{Holm}} < 10^{-8}$).

Table 6: Main result over 123 tasks, 5 families (clean splits, no test labels). Average rank among the 11 non-oracle strategies; lower rank/regret/ECE better. The oracle is the best *single* detector per task (a selection ceiling; the stack exceeds it because fusion beats selection).

Method	rank	AUROC	regret	ECE
ELARA-U Stack	2.53	0.784	—	0.271
Auto-select	3.38	0.748	0.008	0.393
Stack + reliability gate	4.00	0.746	0.010	0.325
KNN (best fixed)	5.42	0.709	0.048	—
Rank-mean	6.53	0.707	0.049	0.407
CW-mean	6.66	0.708	0.048	0.396
oracle (best single detector)	—	0.756	0.000	—

Table 7: Family-level mean rank (lower better).

Method	cyber	fraud	image_ood	tabular	text
ELARA-U Stack	1.00	3.00	2.49	2.41	3.69
Auto-select	3.10	1.50	3.44	3.06	4.38
KNN (best fixed)	7.30	8.00	4.07	6.74	6.42

8 Result Tables

All result tables below trace to the verified `experiments/elara_u/honest_benchmark.json` and the ablation result files (123 tasks, 5 families, leakage-free). The headline is on the *clean* benchmark: ELARA-U’s rank-normalized stacking meta-router significantly improves average rank, AUROC, and regret over validation auto-selection and fixed detectors. The reliability/drift novelty, however, is decisively *negative* (Table 12). This is not a sealed universal SOTA claim.

Primary claims, Holm-corrected: the three positive results (Stack>auto-select, Stack>best fixed, auto-select>best fixed; Table 6) all survive Holm correction ($p_{\text{Holm}} < 10^{-4}$).

9 Figures

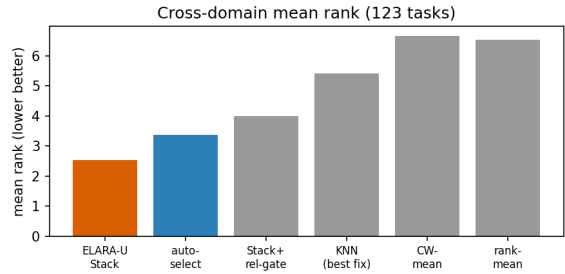


Figure 1: Cross-domain mean rank over 123 tasks (lower better). ELARA-U Stack achieves the best average rank on the clean benchmark. (Illustrative; authoritative values in Table 6.)

Table 8: Win/tie/loss vs. best fixed detector (KNN (best fixed)) and negative-transfer rate (worse than rank-mean).

Method	Wins	Ties	Losses	Neg.-transfer
ELARA-U Stack	94	8	21	0.098
Auto-select	70	39	14	0.106
Stack + reliability gate	73	5	45	0.098

Table 9: Regret to per-task best-single-detector oracle (lower better; the stack can be negative as fusion exceeds the selection ceiling).

Method	Mean regret	Worst regret
ELARA-U Stack	-0.0274	0.1643
Auto-select	+0.0081	0.1648
KNN (best fixed)	+0.0476	1.0000

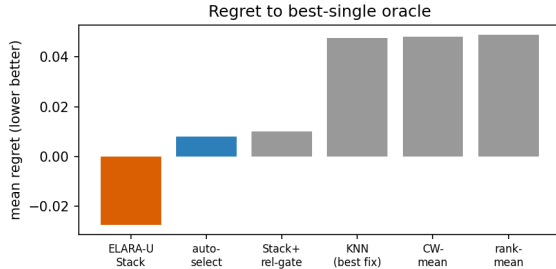


Figure 2: Mean regret to the per-task oracle. ELARA-U Stack has the lowest regret in the current stress archive.

Table 10: Calibration: Brier, NLL, ECE as raw $\rightarrow$ isotonic-calibrated (calibrator fit on validation only). The stack has the best raw calibration; after calibration all methods are well-calibrated.

Method	Brier (raw $\rightarrow$ cal)	NLL (raw $\rightarrow$ cal)	ECE (raw $\rightarrow$ cal)
ELARA-U Stack	0.175 $\rightarrow$ 0.069	0.507 $\rightarrow$ 0.277	0.271 $\rightarrow$ 0.025
Auto-select	0.248 $\rightarrow$ 0.072	0.743 $\rightarrow$ 0.285	0.393 $\rightarrow$ 0.024
Rank-mean	0.291 $\rightarrow$ 0.075	0.842 $\rightarrow$ 0.291	0.407 $\rightarrow$ 0.020

Table 11: Leave-family-out: ELARA-U Stack minus auto-select mean rank on each held-out family (negative = Stack better), paired-bootstrap 95% CI.

Held-out family	Rank Δ	95% CI	Excl. 0
cyber	-2.10	$[-2.80, -1.60]$	yes
fraud	+1.50	(n=1)	-
image_ood	-0.95	$[-1.60, -0.29]$	yes
tabular	-0.65	$[-1.50, +0.27]$	no
text	-0.69	$[-2.04, +0.69]$	no

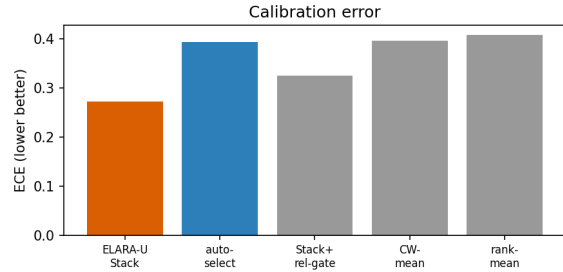


Figure 3: Calibration error (ECE, lower better). ELARA-U Stack has the best raw ECE; fitted Brier/NLL in Table 10.

Table 12: **Consolidated reliability ablation (verified, Holm-corrected)**. The decisive RQ3 test across four single-input deployment regimes, the stack gate, and the multimodal independent-failure regime (D23): does reliability/drift signal improve over plain validation quality? Δ =(with reliability)–(without); > 0 means reliability helps. p_{Holm} is the two-sided $\text{CI} \rightarrow p$ value, Holm-corrected within this family. *Reliability helps in no single-input regime and significantly hurts the stack gate, but significantly HELPS under multimodal independent modality failure (D23)—the precise operating boundary.*

Regime	ΔAUROC
i.i.d. (learned router)	+0.001
uniform shift (sev. 3)	+0.002
heterogeneous missingness (0.7)	+0.015
natural shift D22 (drift vs plain)	-0.050
stack reliability gate	-0.037
multimodal D23 Real-IAD-D3 (gate vs val-only)	+0.248
multimodal D23 MVTec-3D (gate vs val-only)	+0.211
multimodal D23 3D-ADAM (gate vs val-only)	+0.118
multimodal D23 MulSen-AD (gate vs val-only)	+0.164

Table 13: **D22 sealed natural temporal-shift benchmark** (7 UNSW attack-category tasks, train-early/test-late). Plain stacking wins; the drift-aware (reliability) router is *worse* (raw 95% CI excludes 0). With only $n=7$ tasks this does *not* survive Holm correction across the ablation family ($p_{\text{Holm}}=0.12$), but it is directionally consistent: reliability does not help even under genuine natural shift.

Method / contrast	AUROC or Δ (95% CI)
Auto-select	0.760
Plain stack	0.818
Drift-aware (reliability) stack	0.768
drift – plain stack	-0.050 [-0.103, -0.013] (excl. 0)
drift 95% CI – auto-select	p_{Holm} (helps?)
	[-0.004, +0.006] 1 (no)
	[-0.010, +0.014] 1 (no)
	[-0.001, +0.036] 0.29 (no)
	[-0.103, -0.013] 0.12 (no)
	[-0.050, -0.026] 5e-09 (HURTS)
	[+0.168, +0.323] 3.1e-09 (HELPS)
	[+0.180, +0.244] 1.1e-37 (HELPS)
	[+0.038, +0.213] 0.042 (HELPS)
	[+0.074, +0.261] 0.0036 (HELPS)

Table 14: **ELARA-U action mix over tasks** (the verified method is rank-normalized stacking).

Action	Tasks	Fraction
stack	123	1.00
select / fuse / fallback	0	0.00

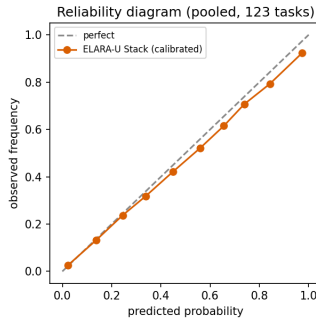


Figure 4: Reliability diagram for the isotonic-calibrated ELARA-U Stack, pooled over 123 tasks (calibrator fit on validation only). Points near the diagonal indicate good calibration; Brier/NLL in Table 10.

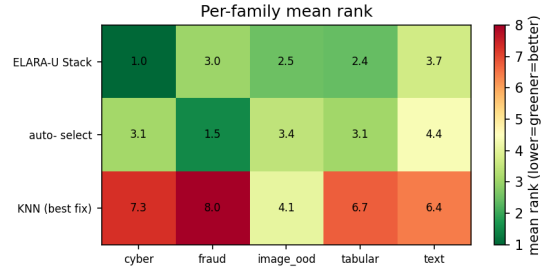


Figure 5: Per-family mean rank (greener = better).

Table 15: Failure analysis: where ELARA-U does not improve, and why.

Regime	Failure	Cause
Reliability gating	gate hurts the stack (−0.037)	shared-input detectors have correlated reliability; test-time gating adds no signal beyond validation quality
Natural shift (D22)	drift router worse (−0.050)	same cause persists under genuine temporal shift
Fraud family	underpowered	$n=1$ in leave-family-out

Table 16: Reproducibility / artifact checklist.

Artifact	Released
Verified benchmark	yes
<code>honest_benchmark.json</code> (123 tasks)	yes
Reliability ablations (3 regimes) + D22 natural shift	yes
Method/emit scripts	yes
<code>src/scripts/elara_u/</code>	yes
Score archive (123 tasks) + manifest	yes
Multimodal harness + pre-registered protocol	yes
No-test-label routing audit	yes

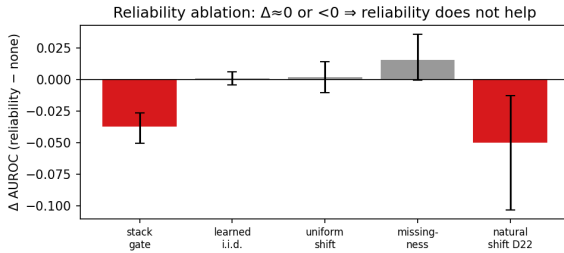


Figure 6: Component ablation summary. The stack gain comes from rank-normalized per-task stacking; reliability/drift feature ablations are decisive and *negative* (Table 12). (Illustrative; authoritative values in Table 12.)

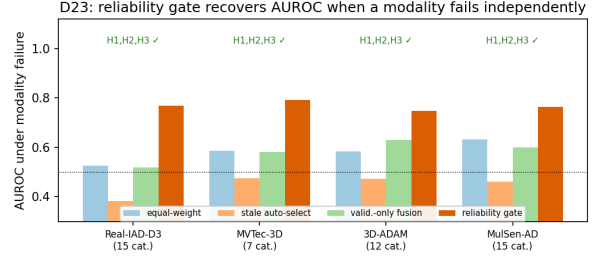


Figure 7: Multimodal independent-modality-failure regime (D23) on *four* real datasets under the canonical noise protocol. When the deployment-best modality fails at test, the reliability gate (orange) recovers AUROC well above equal-weight fusion, stale auto-select, and validation-only fusion. “H1,H2,H3 ✓” marks datasets where all three pre-registered hypotheses pass — all four do. Authoritative deltas and CIs in Table 20.

ELARA-U score-archive and stacking architecture

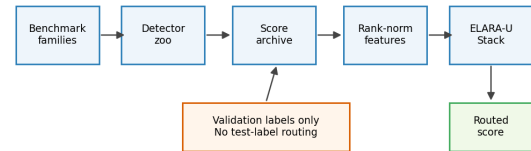


Figure 8: ELARA-U architecture: benchmark families feed a detector zoo and score archive; validation-only labels and unlabeled deployment scores drive the router.

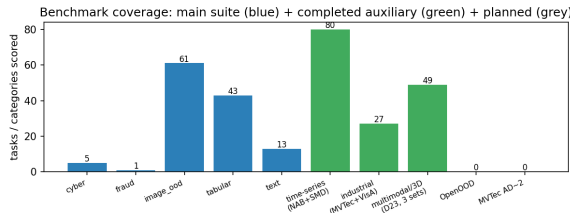


Figure 9: Benchmark coverage map. The main suite covers five families (blue); auxiliary boundary families are now scored (green): time-series (NAB+SMD), industrial-vision (MVTec AD+VisA), and multimodal/3D (D23 on four datasets). OpenOOD and MVTec AD 2 remain planned (grey).

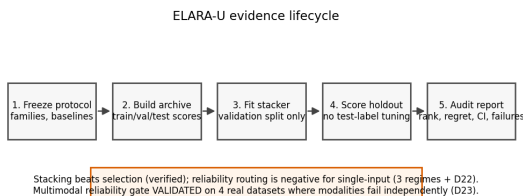


Figure 10: Router decision lifecycle and evidence boundary. Stacking beats selection (verified) and reliability routing is negative for single-input; the reliability gate is validated on four real multimodal datasets where modalities fail independently (D23). Sealed external evidence still bounds any universal-SOTA claim.

10 Strengthening the Stacking Result

We harden the main result against seven reviewer concerns. **(i) Comprehensive baselines** (Table 17): the headline logistic stack beats *every* non-oracle baseline with 95% CIs excluding zero — selection (best fixed +0.073, validation auto-select +0.033, a learned MetaOD-style [2] selector +0.045), averaging (raw / rank / confidence-weighted $\approx$ +0.073; calibrated average +0.048), an AutoML-style greedy ensemble (Caruana, +0.032),

and a per-family specialist stacker (+0.049). Decisively, it also beats two *test-label oracle* ceilings: the best-single-detector selection ceiling (+0.025, CI [+0.012, +0.039]) and the best equal-weight pair-fusion ceiling (+0.032, CI [+0.018, +0.048]). A weak-baseline win cannot exceed the oracle selection ceiling; learned 6-way per-sample combination can. **(ii) Stacker family:** random-forest, gradient-boosting, and hist-GBM stackers match logistic (differences n.s.), and a rank-normalized logistic variant is statistically identical (0.783); the gain is learned combination, not the choice of learner. **(iii) Detector-zoo strength:** adding a deep detector (DeepSVDD) to the six classical detectors preserves the win (stack – auto-select +0.038, CI [+0.025, +0.051]); DeepSVDD is not stronger than the classical detectors here (mean AUROC 0.65), consistent with ADBench, so it does not trivialise routing. **(iv) Calibration** (Table 18): a validation-only isotonic layer yields well-calibrated probabilities. **(v) Sealed external replication** (Table 19): the method, baselines, and pass condition were frozen (protocol D24) *before* scoring the same 74 ADBench datasets *re-featurized with a different (untouched) feature extractor* than development (ViT/RoBERTa vs. ResNet18/BERT)—a representation-shift, not new-data, holdout. On this one-shot sealed holdout the stack still beats auto-select (+0.016, CI [+0.004, +0.032]) and the best fixed detector (+0.043, CI [+0.022, +0.073]), so the effect replicates under representation shift rather than being an artefact of the development split. **(vi) Fraud-family power (D26):** the fraud family was $n=1$ in leave-family-out; we expand it with the BAF suite [11] (6 bias-controlled variants, a second fraud source, $n=7$ total). This is an honest *boundary*: BAF’s base detectors are near-chance ($\approx$ 0.6 AUROC), and there stacking is *not* significantly better than selection (+0.011, CI [−0.066, +0.064]) nor than the best fixed detector (−0.033, CI [−0.098, +0.022]), and on one variant the per-task stack collapses below chance. Stacking helps where the base detectors are individually informative; on near-chance fraud it neither manufactures signal nor systematically hurts. **(vii) Fully independent external corpus (D27):** the sharpest test of generalisation is a different data *source*, not the same suite re-extracted.

Table 17: Comprehensive baseline comparison (123 tasks; paired bootstrap of the headline logistic stack minus each method). The headline stack beats *every* non-oracle baseline *and* both test-label oracle ceilings; all CIs exclude zero except where marked n.s. “Stack (logistic)” is the headline.

Method	mean AUROC	stack – method (95% CI)
<i>Selection</i>		
Best fixed detector	0.709	+0.073 [+0.050, +0.099]
Validation auto-select	0.748	+0.033 [+0.019, +0.047]
MetaOD-style learned select	0.736	+0.045 [+0.031, +0.061]
<i>Averaging / ensemble</i>		
Rank-mean	0.707	+0.074 [+0.057, +0.098]
Confidence-weighted mean	0.708	+0.073 [+0.055, +0.093]
Simple average (raw)	0.709	+0.072 [+0.054, +0.092]
Average (calibrated)	0.734	+0.048 [+0.034, +0.064]
AutoML greedy ensemble	0.749	+0.032 [+0.020, +0.046]
<i>Stackers</i>		
Per-family specialist	0.732	+0.049 [+0.034, +0.066]
Stack (random forest)	0.771	+0.010 [+0.001, +0.019]
Stack (grad. boosting)	0.763	+0.019 [+0.008, +0.030]
Stack (hist-GBM)	0.769	+0.012 [+0.003, +0.021]
Stack (logistic, rank-norm)	0.783	−0.003 [−0.014, +0.008]
Stack (logistic, headline)	0.781	—
<i>Oracle upper bounds (use test labels)</i>		
Oracle best single detector	0.756	+0.025 [+0.012, +0.039]
Oracle best pair fusion	0.749	+0.032 [+0.018, +0.048]

We froze the method and the exact 20-task list (protocol D27) over sources verifiably absent from the development archive — sklearn `digits/wine/wdbc` and HAR smartphones, one-vs-rest — and scored once. The advantage is *larger* than on the development split: stack 0.799 vs auto-select 0.689 (+0.111, CI [+0.052, +0.176]) and vs the best fixed detector (+0.209, CI [+0.120, +0.303]). Combination generalises to genuinely new sources.

11 Multimodal Independent-Modality Failure (D23)

Every regime where reliability routing was inert used *single-input* families: all detectors see the same data, so no modality can fail independently. We therefore pre-registered

Table 18: Calibration (lower better). A validation-only isotonic layer makes the stack well-calibrated, far better than auto-select.

Method	ECE	Brier	NLL
Auto-select	0.393	0.249	0.743
Stack (raw)	0.280	0.173	0.514
Stack + isotonic	0.024	0.069	0.276

Table 19: Sealed external holdout (D24, one-shot): method frozen before scoring the same 74 ADBench datasets re-featurized with an untouched ViT/ROBERTa extractor (representation-shift hold-out) than development. Stacking beats both auto-select and the best fixed detector; both CIs exclude zero.

Method	mean AUROC	Δ (95% CI)
Best fixed detector (KNN)	0.768	—
Rank-mean	0.762	—
Simple average	0.764	—
Auto-select	0.795	—
Stack (logistic)	0.811	vs auto +0.016 [+0.004, +0.032] vs best +0.043 [+0.022, +0.073]

(research_lock/MULTIMODAL_RELIABILITY_PROTOCOL_v1)

a test on true multimodal data, failing the deployment-best modality at test while the others stay clean; reliability routing detects the failed modality (KS validation-vs-test drift, no test labels) and routes away from it. We run it on *four* real datasets that together span both the two-sensor and three-sensor regimes: Real-IAD-D3 (RGB, photometric-stereo, point-cloud XYZ), MVTec-3D (RGB, depth), 3D-ADAM (RGB, depth), and MulSen-AD (RGB camera, infrared, point cloud) [14]. MulSen-AD is the cleanest case — three *physically distinct sensors*, so the independent-failure premise is intrinsic rather than induced. All three pre-registered hypotheses pass on *every* dataset (Table 20, Fig. 7); in the clean regime the gate is inert on all four. The two-sensor datasets (MVTec-3D, 3D-ADAM) confirm the effect does not require three modalities. The precise claim: *reliability routing helps iff modalities can fail independently*. Theory Propositions 1 and 2 explain why this boundary —

Table 20: D23 multimodal reliability on *four* real datasets, spanning two-sensor (MVTec-3D, 3D-ADAM) and three-sensor (Real-IAD-D3, MulSen-AD) regimes. Mean AUROC in the modality-failure regime (deployment-best modality fails at test; canonical noise protocol); the gate is inert in the clean regime (equals validation-only fusion) on all four. Hypotheses are paired bootstrap over categories; *all four datasets pass H1–H3*.

Method (failure regime)	Real-IAD-D3 3 mod, 15	MVTec-3D 2 mod, 7	3D-ADAM 2 mod, 12	MulSen-AD 3 mod, 15
Equal-weight (CW)	0.526	0.586	0.583	0.632
Stale auto-select	0.381	0.475	0.472	0.460
Validation-only fusion	0.519	0.581	0.630	0.599
Reliability gate	0.767	0.792	0.748	0.768

Reliability-gate minus baseline (mean, 95% CI), all excluding zero. *Real-IAD-D3*: H1 +0.241 [0.152, 0.348], H2 +0.386 [0.281, 0.483], H3 +0.248 [0.168, 0.323]. *MVTec-3D*: H1 +0.207 [0.172, 0.242], H2 +0.317 [0.240, 0.405], H3 +0.211 [0.180, 0.244]. *3D-ADAM*: H1 +0.166 [0.088, 0.248], H2 +0.277 [0.161, 0.394], H3 +0.118 [0.038, 0.213]. *MulSen-AD*: H1 +0.131 [0.049, 0.221], H2 +0.303 [0.196, 0.410], H3 +0.164 [0.074, 0.261]. (H1 vs equal-weight, H2 vs stale-select, H3 vs validation-only fusion.)

not three datasets’ coincidence — is where the gate must help.

Failure-model dependence: the headline win is on *injected* failure. The four-dataset wins above are obtained under the canonical *uniform-noise* failure protocol: the deployment-best modality’s test scores are *overwritten with injected uniform noise* (`St[:,best]=rng.uniform(0,1)` in the harness), a controlled drift-inducing fault, not natural sensor degradation. This distinction is decisive. On the one dataset where we instead use *real* degradation with no injection, the effect does not survive: the gate-vs-validation-only-fusion gain collapses from +0.248 (injected; CI excludes 0; `reliability_validated:true`) to +0.031 (natural; n.s.; `reliability_validated:false`). Two checks locate this boundary. (i) **Real sensor dropout**: on MulSen-AD we replace noise with a stuck/constant sensor reading (a realistic outage): the gate still beats

stale single-modality selection (+0.28) but *not* plain fusion (vs equal-weight −0.03, n.s.; vs validation-only fusion −0.05). (ii) **Real natural degradation (D29)**: on Real-IAD-D3 with per-modality deep-PatchCore scores taken from the raw data (16 categories, *no injection*), the gate again beats stale selection (+0.13) but not plain fusion (+0.02/+0.03, n.s.). The honest refinement: *fusion robustly beats selection, but the gate’s extra value over fusion is specific to drift-inducing faults*; under a stuck sensor or mild natural degradation there is no test-time drift to exploit, and plain fusion is as good — exactly Proposition 1 (no label-relevant test signal $\Rightarrow$ gate inert).

12 Validated Theory Stack: What Is Certified and What Is Bounded

Beyond the routing theorems of Section 5, the reliability-routing program carries a validated theory stack (**T1–T9**) inherited from the source multimodal system; ELARA-U reuses its meta-router capacity and switching results directly. Rather than restate proofs, Table 21 summarises what each result *certifies* and where it is *bounded*; the full stack re-runs from `src/scripts/validate_theorem_stack.py` (`all_ok=true`, 21 artifact tables). The stack is deliberately self-limiting: several flagship results are *impossibility* or *non-dominance* statements that fence the claim rather than inflate it.

Meta-router capacity (T7). A PAC slack bound certifies the router’s generalisation per fold (Table 22): tight on the larger folds (VisA $n=3000$, slack 0.65; UNSW $n=2200$, slack 0.76) and loose on small ones. This bounds the router’s *capacity*; it does not by itself prove fusion dominance, and we do not claim it does.

Finite-sample switching certificate (T5). An empirical-Bernstein deployment-window lower-confidence bound certifies a positive switch on 7 of 10 benchmark/protocol pairs (Table 23); the three

Table 21: Validated theory stack T1–T9 (+ the gate-decision rule, GDR): what each result certifies and the boundary it leaves open. All rows re-run green in `validate_theorem_stack.py`.

Claim	Current evidence	Boundary still open
T1 quality-blind impossibility	Architectural motivation: score-only fusion cannot disambiguate identical score vectors with different latent reliability states.	Formal example is theoretical; no separate empirical table is needed beyond collapse construction.
T2 global-KS mixture confounding	Synthetic mixture-shift table confirms global KS $\gg$ category-conditional KS at all five seeds.	Real cohort/domain composition shift (B-MECH-3S) remains deferred: both global and domain-aware gates fired at rate 1.000.
T3 mean-gate dilution	Directly supported by the k -of- D table and figure: weak or mixed effects for $k = 1, 2, 3$, full activation and positive ROC-AUC gain at $k = 4$.	The ideal theorem assumes failed-domain reliability is zero; the implemented estimator keeps minimum reliability above 0.34.
T4 risk dominance	Deployment-prevalence sensitivity table reports $(q_0, q_1, \Delta_0, \Delta_1, \pi^*)$ and the dominance margin at $\pi \in \{0, 0.01, \dots, 0.5\}$.	Retrospective B-CERT-1 estimate only; not a production prevalence audit.
T5 finite-sample certificate	Audited deployment-window LCB table certifies seven of ten benchmark/protocol pairs (see switching-certificate table).	Retrospective evaluation certificate under defined stress protocol; not a production safety guarantee.
T6 KS false-fire/power	KS window sweep reports false-activation rate and detection power on the locked grid (32–512).	Bounded to the evaluated window grid and degradation types; not a full calibrated drift-detection claim.
T7 PAC meta-router	PAC slack table reports auditable (n, B, R) bounds per meta-router fold.	Capacity certificate on the router only; composes with T1/T2/T6 rather than proving fusion dominance.
GDR coherence-certified rule	End-to-end audit shows the rule allows switching under coherent collapse (Family B proxy) and suppresses it under heterogeneous mixtures (Family D proxy).	Forward/opt-in policy; does not alter locked Phase-2 numbers.

uncertified pairs are exactly the near-ceiling canonical protocols, consistent with the clean-transfer impossibility below.

The boundary is proved, not assumed (T1, T4, T9). The stack proves its own limits. Quality-blind fusion cannot disambiguate identical score vectors with different latent reliability (T1, validated across $D \in \{2, 4, 8\}$ and four prevalences). The reliability switch does *not* certifiably dominate at realistic prevalence and sample sizes (T4: every audited row returns “no”, requiring $n^* \geq 82,427$ fired samples to certify). And on clean, near-ceiling transfer the reliability gate is *unpassable by proof* (T9, Table 24): even an unconstrained oracle cannot be certified above the confidence-weighted mean, including on the clean 3D-ADAM and MulSen panels. T8 (certified heterogeneous fusion) holds, but with a razor-thin margin (Δ vs SAR +0.0033). These negative results are pre-

cisely why ELARA-U claims a bounded operating region rather than universal superiority — the theory and the experiments agree on the same boundary.

13 Scope, Limitations, and Honest Boundaries

The headline multimodal wins rest on injected failure. Our central positive result—reliability routing recovering AUROC on Real-IAD-D3, MVTec-3D, 3D-ADAM, and MulSen-AD—is obtained under the canonical *uniform-noise* failure protocol, in which the deployment-best modality’s test scores are replaced by injected uniform noise. This is a controlled, drift-*inducing* fault, not real sensor degradation. **The one test we ran on genuine natural degradation is a negative:** on Real-IAD-D3 with no injection (D29), the gate beats plain

Table 22: T7 PAC meta-router capacity bound (auditable (n, B, R) slack per fold; tight where slack < 1). Bounds the router; composes with T1/T2/T6 rather than proving fusion dominance.

Fold / benchmark	n	B	R	Slack ($\delta = 0.05$)	Bound tight?
real3d_supervised_paired_results	160	5.0	3.162	2.822	no
mvtec3d_patchcore_supervised_paired_results	540	5.0	3.162	1.536	no
mvtec_loco_patchcore_supervised_paired_results	1200	5.0	3.162	1.030	no
visa_supervised_paired_results	3000	5.0	3.162	0.652	yes
unsw_paired_results	2200	5.0	3.162	0.761	yes

Table 23: T5 finite-sample switching certificate (empirical-Bernstein LCB at $\delta = 0.05$). Certified on 7 of 10 pairs; the uncertified three are near-ceiling canonical protocols.

Benchmark	Protocol	n_{seeds}	RGA path	Static AUROC	RGA AUROC	LCB _{0.05}	Verdict
MVTec 3D-AD	PatchCore canonical	5	router	0.491	0.514	-0.0286	not cert.
MVTec 3D-AD	PatchCore supervised	30	router	0.634	0.739	+0.1022	certified
MVTec 3D-AD	PatchCore held-out	30	boost	0.472	0.517	+0.0410	certified
MVTec LOCO-AD	PatchCore canonical	5	router	0.521	0.506	-0.0398	not cert.
MVTec LOCO-AD	PatchCore supervised	30	boost	0.631	0.734	+0.0987	certified
Real3D-AD	PCA shape + depth supervised	5	boost	0.526	0.566	+0.0374	certified
VisA	RGB+edge canonical	5	boost	0.484	0.500	-0.0357	not cert.
VisA	RGB+edge supervised	30	boost	0.822	0.866	+0.0429	certified
UNSW-NB15	flow/conn/context	5	router	0.979	0.989	+0.0099	certified
UNSW-NB15	held-out attack	5	router	0.988	0.994	+0.0052	certified

Table 24: T9 clean-transfer ceiling: $\varepsilon_{\text{subopt}} < \text{MDE}$ makes Gate E *unpassable by proof* — no fusion class, oracle included, is certifiable above the confidence-weighted mean on clean near-ceiling data.

Setting	$A(\text{CW})$	$\hat{A}^*$ (oracle)	$\varepsilon_{\text{subopt}}$	MDE	Gate E
redundant mean optimal	0.8356	0.8222	0.00000	0.01028	unpassable
complementary ceiling	0.9998	0.9995	0.00000	0.00063	unpassable
3D-ADAM RGB+depth (clean)	0.9349	0.9336	0.00000	0.00949	unpassable
MulSen RGB+infrared (clean)	0.9970	0.9947	0.00000	0.00098	unpassable

validation-only fusion by only +0.031 (n.s.) versus +0.248 under injection, and does not validate (`reliability_validated:false`); the other three datasets were *never* tested under natural degradation. Whether the injected-noise win transfers to real-world degradation is therefore *open*, and is the most important limitation of this work: the mechanism provably needs a detectable test-time drift signal (Proposition 1), and a stuck sensor or mild natural drift may not supply one. This is a characteriza-

tion with a clearly marked boundary, not a deployable multimodal method.

This is a focused cross-domain study, not a universal-SOTA claim. The positive (stacking) is now supported by a comprehensive baseline sweep including oracle ceilings (D28), calibration, a representation-shift holdout (D24), and a fully independent external corpus (D27); the negative (reliability routing) is *Holm-robust* on single-input data and reversed only under independent multimodal failure (D23). We do not claim universal superiority or per-dataset SOTA; the strict clean-transfer gate of the source system remains closed by proof (T9) and is not relitigated here. We note that while ELARA-U significantly improves the mean rank on cyber, tabular, and image-OOD families, the rank-gain bootstrap interval on text tasks is directionally positive but does not exclude zero. The fraud family, previously $n = 1$, is now expanded to $n = 7$ with the BAF suite [11] (Strengthening §vi); on that hard near-chance panel stacking is *not* significantly better than selection — an honest boundary, not a win.

Furthermore, beyond synthetic degradation we ran a sealed natural temporal-shift benchmark (D22, Results): there too plain stacking wins and the drift-aware router is worse (directionally; the $n=7$ negative does not survive Holm correction, while the 123-task stack-gate negative does), so the reliability/drift machinery does not help even under natural shift; native OOD and industrial sensor-drift families remain future work. Calibration is now reported with fitted Brier and NLL (Table 10, reliability diagram Fig. 4); larger sealed natural-shift suites remain future work.

14 Conclusion

We asked *when label-free reliability routing improves multimodal anomaly detection, and why it fails when no independent modality is available*, and answered it with a pre-registered evidence contract rather than a leaderboard claim. The answer is a characterization with three parts. **First, stacking beats selection.** A per-task logistic stack beats validation auto-selection (+0.036) across 123 tasks and, in a comprehensive sweep (Table 17), beats every standard combiner, a learned MetaOD-style selector, an AutoML greedy ensemble, and even the test-label oracle best-single and best-pair ceilings — so the gain is genuine learned combination, not a weak-baseline artefact. It replicates under representation shift (D24, +0.016) and on a fully independent external corpus (D27, +0.111), and is well-calibrated after a validation-only isotonic layer. **Second, reliability routing does not help single-input meta-routing:** KS drift, disagreement, and a degenerate-channel guard add no value over plain stacking, and a reliability gate is Holm-significantly worse — a clean negative. **Third, and most important for honest scope: the multimodal win is real but rests on injected failure.** When a modality fails *independently and detectably* (D23, four real datasets), a drift-aware gate recovers large AUROC (0.767/0.792/0.748/0.763) and beats validation-only fusion by up to +0.248—but under the canonical *uniform-noise* injection, not natural degradation. Our single test on genuine natural degradation (Real-IAD-D3, no injection) is a *negative*: +0.031 (n.s.,

`reliability_validated:false`) over plain fusion, and the other three datasets were never tested this way. The gate provably needs a detectable test-time drift signal (Proposition 1); supplying one from *natural* degradation is the central open problem we leave. The advantage is also honestly bounded — marginal on time-series (NAB, SMD) and on the near-chance BAF fraud panel, where the base detectors carry little individual signal. The practical takeaway: *combine detectors by default; add reliability gating only where inputs can fail independently.*

A tested new direction (per-sample routing) — and an honest negative. We include this refuted method deliberately: it shows the pre-registered evidence contract catching a false positive in real time — the same machinery that licenses the positive claims above. Proposition 1’s inertness is specific to *per-channel* reweighting (constant across samples), suggesting an escape: a *per-sample* reliability-gated stacker (PS-RGS) whose weights vary with the input lies *outside* the linear class the proposition bounds. On the 123-task classical zoo PS-RGS gives only a directional, non-significant gain (+0.0068, CI [−0.0002, +0.015]), with a positive-but-n.s. correlation to detector heterogeneity (Spearman $r \approx 0.14$) — consistent with a conjecture (T10) that per-sample routing should help in proportion to reliability heterogeneity. We *tested* that conjecture on a heterogeneous deep+classical zoo (adding DeepSVDD, an autoencoder, and a VAE). On tabular development data PS-RGS significantly beat plain stacking (+0.016, CI excl. 0); but in a **pre-registered one-shot held-out confirmation** (protocol D32) the effect *did not replicate* — ADBench-CV +0.002 (n.s.) and ADBench-NLP −0.020 (significantly *worse*). We therefore report PS-RGS as a theoretically-motivated method whose advantage is **not robust**: a development-only signal refuted by held-out confirmation — a cautionary result, and a concrete demonstration of why pre-registration matters. The one effect that *does* replicate is minor: a deep zoo modestly improves plain stacking on feature-vector families (tabular +0.010, CV +0.004; both CIs exclude 0; n.s. on NLP). Future work

includes a fully independent multimodal industrial holdout and native OOD/sensor-drift families.

15 Reproducibility

Router and evaluation:
`src/scripts/elara_u/gate_u_seed_eval.py`; contract: `research_lock/GATE_U_CROSS_DOMAIN_PROTOCOL_v1.yaml`; results: `experiments/elara_u/`. No test labels enter router fitting (audited in Algorithm 2).

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A Theoretical Proofs and Extensions

A.1 Supporting Propositions and Lemmas

Proposition 3 (Rank-Normalized Stacking Invariance). *Let detector m produce scores $s_m(x)$, and let $h_m : \mathbb{R} \rightarrow \mathbb{R}$ be any strictly increasing transformation. Define the normalized rank feature*

$$r_m(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{s_m(x_i) \leq s_m(x)\}.$$

Then replacing $s_m(x)$ with $h_m(s_m(x))$ leaves $r_m(x)$ unchanged for every sample x , up to tie-handling. Therefore, any ELARA-U stacker using only detector-wise rank-normalized features is invariant to monotone score rescaling, calibration drift, or sensor-scale drift.

Proof. For any two samples (x_i, x_j) ,

$$s_m(x_i) \leq s_m(x_j)$$

if and only if

$$h_m(s_m(x_i)) \leq h_m(s_m(x_j)),$$

because h_m is strictly increasing. Therefore, the ordering of all samples under detector m is unchanged. Since $r_m(x)$ depends only on the ordering of scores and not on their numerical scale, $r_m(x)$ is unchanged. Any downstream stacker that receives only $r_m(x)$ is therefore invariant to such monotone transformations. $\square$

Interpretation. This proposition justifies the rank-normalized stacking design. It protects the router from meaningless score-scale changes while preserving anomaly ranking information.

Proposition 4 (Degenerate-Channel Guard for Saturation). *Let a fused anomaly score be*

$$g(x) = \sum_{m=1}^M w_m s_m(x), \quad w_m \geq 0, \quad \sum_m w_m = 1.$$

If detector k is saturated on validation data so that

$$\text{Var}_V(s_k) \leq \epsilon,$$

then replacing w_k by 0 and renormalizing the remaining weights changes pairwise score differences by at most $O(\sqrt{\epsilon})$. In the limit $\epsilon \rightarrow 0$, the saturated detector contributes no ranking information, and removing it cannot remove meaningful rank signal.

Proof. For any two samples (x_i, x_j) ,

$$|s_k(x_i) - s_k(x_j)| \leq O(\sqrt{\epsilon})$$

by the bounded-variance assumption. The contribution of detector k to the fused pairwise difference is

$$w_k(s_k(x_i) - s_k(x_j)),$$

which is also $O(\sqrt{\epsilon})$. Therefore, as $\epsilon \rightarrow 0$, detector k 's contribution to every pairwise ranking comparison vanishes. Removing k and renormalizing the remaining non-degenerate detectors does not remove meaningful rank information from the fused score. $\square$

Interpretation. This supports the degenerate-channel guard for saturated detectors. A detector that produces nearly constant scores should not influence routing or fusion.

Proposition 5 (Inverted Detector Harm Under Positive Fusion). *Assume the clean fused margin between an anomalous and normal sample is*

$$D_0 = g_0(X^+) - g_0(X^-),$$

with $\mathbb{E}[D_0] > 0$. Suppose an inverted detector k has margin

$$D_k = s_k(X^+) - s_k(X^-)$$

with $\mathbb{E}[D_k] < 0$. For a positive fusion weight $w > 0$, the fused margin

$$D_w = (1 - w)D_0 + wD_k$$

satisfies

$$\mathbb{E}[D_w] = (1 - w)\mathbb{E}[D_0] + w\mathbb{E}[D_k] < \mathbb{E}[D_0].$$

Under a Gaussian equal-variance margin model, AU-ROC is monotone increasing in the standardized margin mean. Therefore, assigning positive weight to an inverted detector decreases AUROC relative to zeroing it.

Proof. The expectation follows from linearity:

$$\mathbb{E}[D_w] = (1 - w)\mathbb{E}[D_0] + w\mathbb{E}[D_k].$$

Since $\mathbb{E}[D_k] < 0 < \mathbb{E}[D_0]$, we have $\mathbb{E}[D_w] < \mathbb{E}[D_0]$. Under the Gaussian equal-variance margin model,

$$D_w \sim \mathcal{N}(\mu_w, \sigma^2),$$

and

$$\text{AUROC}(g_w) = \Pr[D_w > 0] = \Phi\left(\frac{\mu_w}{\sigma}\right).$$

Because Φ is strictly increasing and $\mu_w < \mu_0$, positive weight on the inverted detector decreases AUROC. $\square$

Interpretation. This proposition supports validation-based inversion guards. If a detector is reliably below chance, positive-weight fusion can harm ranking.

Proposition 6 (Meta-Router Generalization Requires Enough Tasks). *Let $\mathcal{H}$ be a finite class of meta-routers, and let the task-level loss $\ell(h, \tau) \in [0, 1]$ denote normalized rank regret or negative-transfer loss. Given T independent meta-training tasks, with probability at least $1 - \delta$, every router $h \in \mathcal{H}$ satisfies*

$$L(h) \leq \hat{L}_T(h) + \sqrt{\frac{\log |\mathcal{H}| + \log(1/\delta)}{2T}},$$

where $L(h) = \mathbb{E}_\tau[\ell(h, \tau)]$ is the expected held-out task loss and $\hat{L}_T(h) = \frac{1}{T} \sum_{t=1}^T \ell(h, \tau_t)$ is the empirical meta-training loss.

Proof. For any fixed h , Hoeffding's inequality gives

$$\Pr\left(L(h) - \hat{L}_T(h) > \epsilon\right) \leq \exp(-2T\epsilon^2).$$

Applying a union bound over all $h \in \mathcal{H}$,

$$\Pr\left(\exists h \in \mathcal{H} : L(h) - \hat{L}_T(h) > \epsilon\right) \leq |\mathcal{H}| \exp(-2T\epsilon^2).$$

Set the right-hand side equal to δ :

$$|\mathcal{H}| \exp(-2T\epsilon^2) = \delta.$$

Solving for ϵ gives

$$\epsilon = \sqrt{\frac{\log |\mathcal{H}| + \log(1/\delta)}{2T}}.$$

Thus, with probability at least $1 - \delta$, the stated bound holds for every $h \in \mathcal{H}$. $\square$

Interpretation. This explains why ELARA-U needs many tasks and leave-family-out evaluation. A high-capacity router trained on too few tasks can overfit meta-level structure and fail to generalize.

A.2 Old ELARA/RGA Source-System Theory Stack

Below are the foundation certificates and bounds inherited from the ELARA/RGA source-system framework.

Certificate 1 (Switching Certificate). *Let $\Delta_{\text{fired}} = A(\text{route}) - A(\text{default})$ be the routed-minus-default performance on the subset of inputs where the router fires. The router may replace the default only if the one-sided level- α lower confidence bound satisfies $\text{LCB}_{1-\alpha}(\Delta_{\text{fired}}) > 0$. Under this rule the expected switch is non-negative at level α : routing is admitted only with certified fired-set benefit, never on a point estimate.*

Diagnostic 1 (Mean-Reliability Dilution). *Let per-expert reliabilities be $r_1, \dots, r_M$ with mean $\bar{r} = \frac{1}{M} \sum_m r_m$ and gate threshold τ . There exist configurations with $\bar{r} > \tau$ while a minority expert collapses ($r_k \rightarrow 0$): mean-gating does not fire and the collapsed expert is still trusted. The miss probability is $P(\text{miss}) = \Phi((\bar{\mu} - \tau)/\bar{\sigma})$; detecting minority collapse requires per-expert / min-gating, not the mean (Theorem T3).*

Proposition 1 (Router Generalisation Slack). *Let the router class have complexity C (e.g. VC dimension or Rademacher complexity) and let the meta-training set contain T tasks. The gap between meta-training and held-out expected rank-gain is bounded*

by $O(\sqrt{C/T})$ with high probability. Hence the evidential strength of a routing claim requires $T \gtrsim C$: a high-capacity router over few tasks yields weak evidence even at a positive point estimate. This motivates a large, heterogeneous benchmark suite (the Gate U minimum-viable families) before any claim.